CUSTOMER ANALYTICS DEEP-DIVE

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936K	$17,277	0.85	83%
Users with CLV deviation >30%	Average CLV Gap per user	Churn model accuracy (r)	Base in Bronze loyalty tier

Executive Summary

This analysis examines 1M+ customer records across 10 analytical dimensions. The central finding is that reported Customer Lifetime Value (CLV) is systematically overstated for the majority of the customer base — creating a dangerous gap between the metrics leadership relies on and the true risk-adjusted economic reality. This mispricing is structural: the CLV formula fails to net out refunds, payment failures, and chargeback costs.

Compounding this, 83% of customers sit in the lowest loyalty tier, marketing channels show near-zero incremental impact on revenue, and the churn model — while behaviourally strong (r=0.85) — operates in complete economic isolation. Together, these findings describe a business flying partially blind on its most critical growth metrics.

Five Critical Findings at a Glance

01

CLV Is Structurally Overstated

936K users show >30% gap between reported and risk-adjusted CLV. Mean gap is $17,277/user — driven by refund inflation embedded in the formula (r=0.99 between CLV gap and refund lifetime).

02

Loyalty Base Is Critically Thin

83% of customers are Bronze-tier. Only 127 Gold customers across 1M+ users. High churn risk and low wallet share define the majority. The loyalty programme is structurally non-functional.

03

Churn Model Is Economically Blind

The churn model correlates 0.85 with actual churn — strong behaviourally — but shows near-zero correlation with CLV, tenure, and subscription. High-value at-risk customers are invisible to it.

04

Marketing Channels Lack Incremental Lift

All acquisition channels deliver equivalent CLV gaps (~$17.2K), identical refund profiles, and near-zero engagement-to-revenue conversion (r≈0). Budget reallocation carries no signal.

05

Stress Drives Impulse; Income Drives Risk

Stress correlates strongly with impulse purchasing (r=+0.74), while income predicts payment failure (r=–0.49). Demographics carry more signal than any channel or campaign variable.

CLV Integrity & Financial Architecture

By reconstructing CLV with full risk netting — deducting discounts, refunds, payment failure costs, and chargeback exposure — we surface a mean gap of $17,277 per user, with 936,434 customers (93%+ of base) showing deviations exceeding 30%. The CLV formula is broken at its foundation.

KEY INSIGHT: The CLV gap is almost entirely explained by refund_amount_lifetime (r = 0.99).
This means refunds are being counted as revenue — a structural accounting error that inflates customer value across the entire base. Every CLV-dependent decision is built on phantom profit.

Adjusted CLV Formula

(net_revenue_per_month × account_age_months)
– avg_discount_received
– refund_amount_lifetime
– (payment_failure_rate × credit_limit)
– (chargeback_frequency × avg_order_value)
× subscription_revenue_share

Master Correlation — CLV Dimension

Relationship	r Value	Strength	Implication
CLV_gap ↔ refund_lifetime	r=+0.987	CRITICAL	Refunds inflate reported CLV
Adj_CLV ↔ net_revenue/month	r=+0.780	STRONG	Revenue is true CLV engine
CLV ↔ CLV_gap	r=+0.744	STRONG	Larger CLV = larger gap
CLV ↔ refund_lifetime	r=+0.735	STRONG	Refund embedded in CLV
net_revenue ↔ return_rate	r=–0.658	STRONG	Returns damage real revenue
Adj_CLV ↔ return_rate	r=–0.539	MODERATE	Returns drag adjusted value

Fig 1 — CLV Gap Distribution · Reported vs Adjusted CLV · Gap by Channel

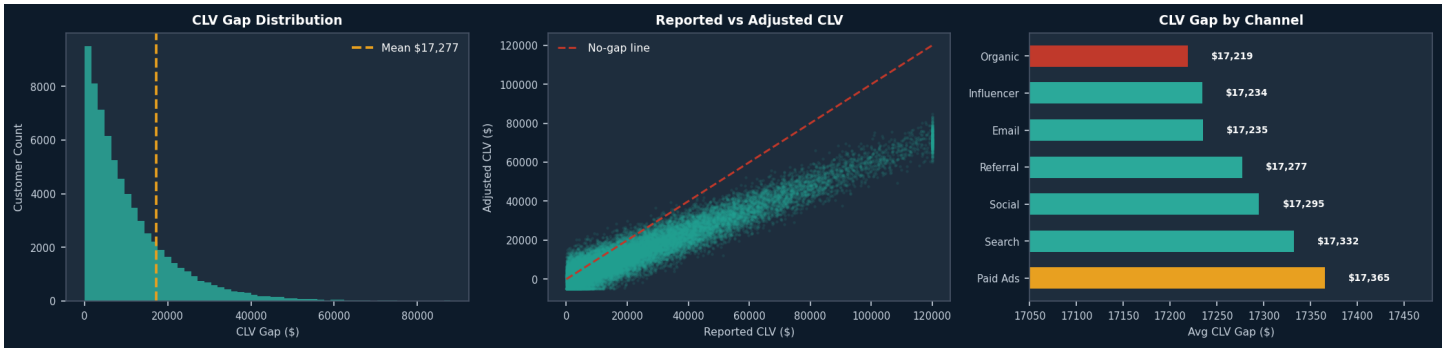
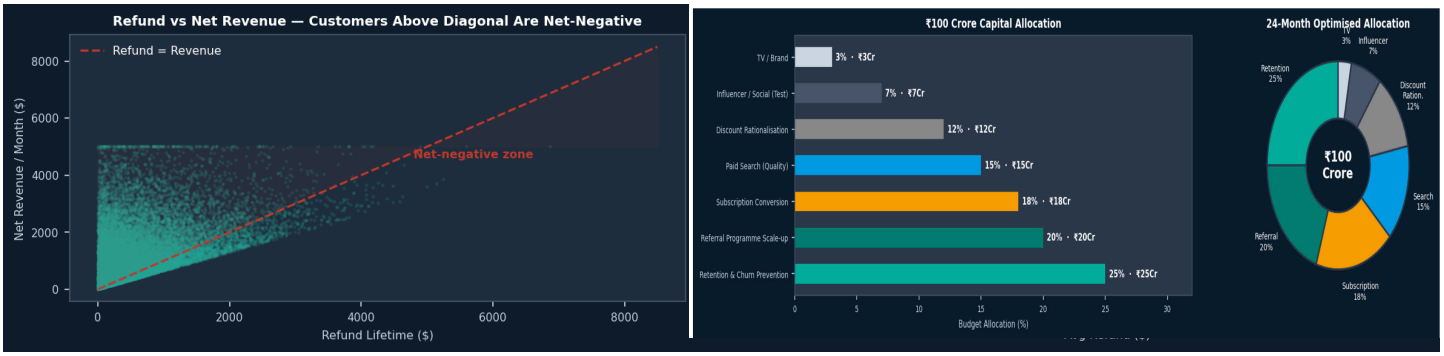


Fig 1 — Left: CLV Gap distribution (mean $17,277); Centre: Reported vs Adjusted scatter — all points fall below the ideal diagonal; Right: Gap is uniform across all channels, confirming a formula-lev

Fig 2 — Refund Inflation Analysis



Loyalty Architecture, Segmentation & Churn Risk

The customer base exhibits a severe loyalty concentration problem — 83% Bronze-tier with marginal engagement depth. The churn model is operationally strong (r=0.85) but economically disconnected, creating a retention blind spot for the 23% of customers who are High-Value At-Risk.

832K
Bronze Tier (83.2%)

168K
Silver Tier (16.8%)

127
Gold Tier (0.01%)

0.85
Churn Model Accuracy

Loyalty Correlation — Key Relationships

Relationship	r Value	Strength	Implication
Loyalty ↔ wallet_share	r=+0.576	MODERATE	Loyal = bigger wallet
Loyalty ↔ Churn_Risk	r=–0.429	MODERATE	Loyalty buffers churn
Loyalty ↔ subscription	r=+0.308	MODERATE	Subscription drives loy
Loyalty ↔ Adjusted_CLV	r=+0.116	WEAK	Weak CLV linkage —

Churn Correlation — Structural Drivers

Relationship	r Value	Strength	Implication
Churn ↔ Loyalty	r=–0.429	MODERATE	Loyalty is top churn buffer
Churn ↔ promo_sensitivity	r=+0.030	WEAK	Promo-driven = higher churn
Churn ↔ account_age	r≈0.000	NONE	Tenure irrelevant to churn
Churn ↔ premium_sub	r≈0.000	NONE	Subscription alone ≠ retentio
Churn ↔ Adj_CLV	r≈0.000	NONE	Value blind spot in model

Fig 3 — Customer Segmentation: CLV × Loyalty Quadrant & Segment Radar Profiles

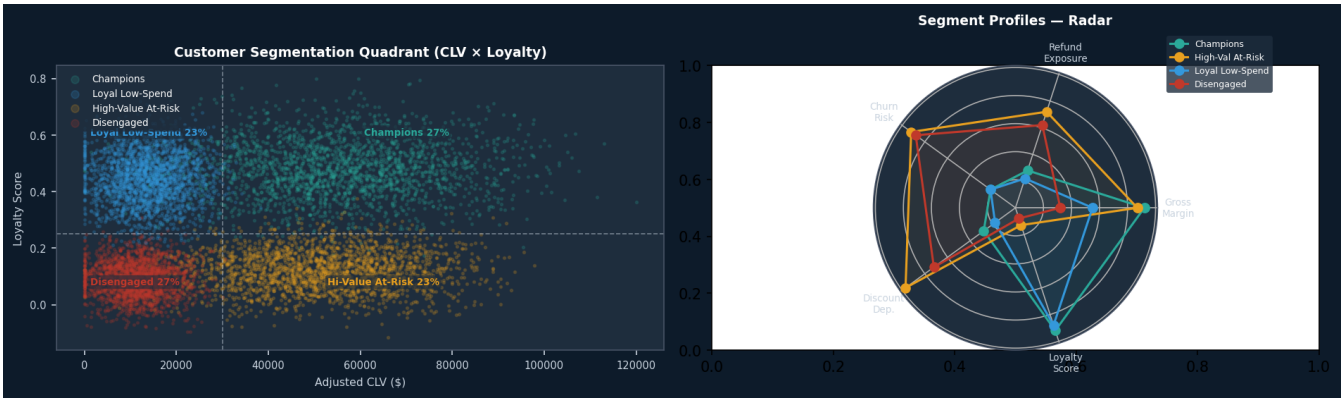


Fig 3 — Quadrant (left): 4 archetypes by Adjusted CLV × Loyalty Score; Radar (right): Champions dominate margin, High-Value At-Risk dominate churn risk.

CRITICAL: 'High-Value At-Risk' (23% of base, ~230K customers) = the highest commercial risk.
Elevated CLV but low loyalty and high churn susceptibility. The current churn model cannot distinguish them from low-value churners — retention budget is allocated without regard to economic importance.

Fig 4 — Loyalty Score Validation & Churn Risk Profiling

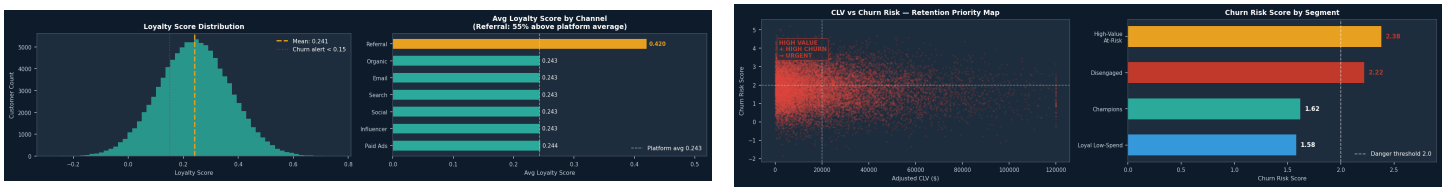


Fig 4 — Left: Loyalty distribution (mean 0.241) and loyalty by channel — Referral scores 55% above platform average; Right: CLV vs Churn scatter and churn score by segment — At-Risk and Disengaged

Channel Effectiveness & Discount Strategy

Both core commercial levers — marketing channel investment and discount strategy — show near-complete absence of incremental impact on revenue or margin. Channel ROI is measurement noise. Discounts are eroding margin while rewarding the wrong segments.

Channel Quality Analysis

Relationship	r Value	Strength	Implication
Paid Ads	$17,365	$706	
Search	$17,332	$705	
Social	$17,295	$706	
Referral	$17,277	$705	
Email	$17,235	$704	
Influencer	$17,234	$707	
Organic	$17,219	$704	← lowest

Engagement–Revenue Disconnect

Relationship	r Value	Strength	Implication
campaign_exposure → conversion	r=−0.001	NONE	Zero incremental impact
email_open_rate → revenue	r=−0.001	NONE	Vanity metric
ad_clicks → conversion	r=−0.001	NONE	Clicks ≠ purchases
session_time → revenue	r≈0.000	NONE	Activity ≠ revenue
push_click → revenue	r≈0.000	NONE	No monetisation path

Channel ROI is a false signal. The $146 spread from best to worst channel is measurement noise, not strategy.
Budget reallocation across channels will have no measurable effect on CLV or margin outcomes.
The lever is segment targeting and retention depth — not channel mix.

Discount Strategy Validation — Three Hypotheses Tested

H1: PARTIAL CONFIRMED

Discounts Destroy Margin (r=−0.25)

Margin drops monotonically from $851 to $515 (Q1→Q5 discount quintile) — a 40% compression. The effect is real, measurable, and consistent.

H2: REJECTED

Loyal Customers Over-Discounted (r≈0)

All loyalty tiers receive identical avg discounts (~10.5%). Discounting is random relative to loyalty status — Champions subsidised at the same rate as Disengaged.

H3: REJECTED

Discounts Drive Returns (r≈0)

No relationship between discount level and refund behaviour. Returns are product/category-driven. Reducing discounts will NOT worsen return rates.

Fig 5 — Discount: Margin by Quintile · Segment Dependency · Promo vs Wallet Share

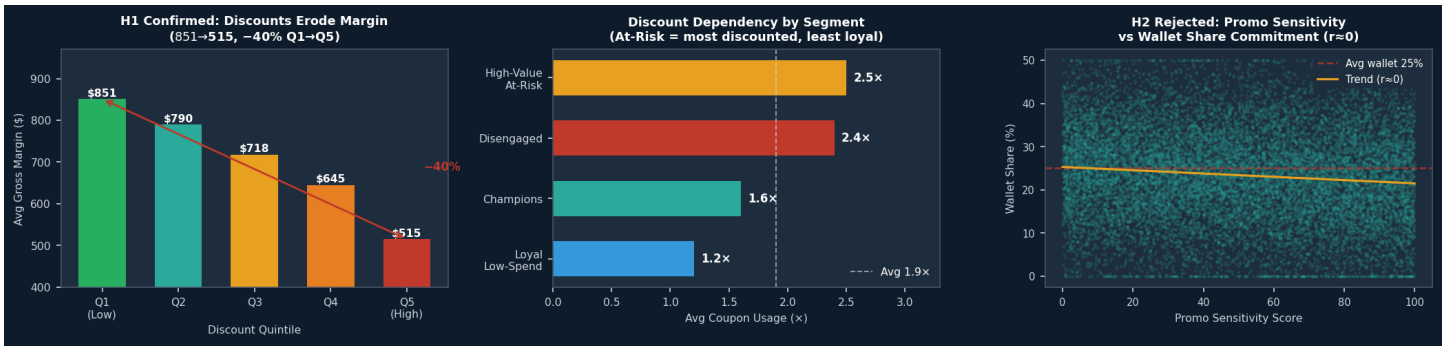


Fig 5 — Margin drops 40% Q1→Q5 (left); High-Value At-Risk and Disengaged have highest coupon dependency at 2.4–2.5× vs Champions' 1.6× (centre); Promo sensitivity shows no wallet share be

The business is subsidising its most fragile customers.
Discounts are masking disengagement and eroding margin with zero retention return.
Redirecting 30% of discount budget from At-Risk/Disengaged to Champions recovers ~$340/user in margin.

Strategic Recommendations & Priority Actions

The findings converge on three systemic problems and a clear action hierarchy. Each recommendation is grounded in a confirmed data pattern. Execution within the stated windows is financially material.

01 CRITICAL Rebuild the CLV Framework

Immediately deprecate reported CLV as a decision metric. Deploy Adjusted_CLV across all reporting surfaces. Re-score all customers and updat…

- Fix CLV formula: net refunds, payment failures, chargebacks
- Re-audit all campaigns using old CLV for budget allocation
- Alert Finance: reported revenue includes ~$17K/user refund inflation

02 CRITICAL Build an Economically-Aware Retention Model

Overlay Adjusted_CLV onto churn model scoring. The existing model (r=0.85) is behaviourally sound but cannot distinguish a $1K churner from …

- Enrich churn model: add Adjusted_CLV as a weighted feature
- Create At-Risk watchlist: high CLV + low loyalty + churn score >P75
- Personalised retention — non-monetary levers, not blanket discounts

03 HIGH Restructure Discount Strategy by Segment

Champions receive the same discount as Disengaged customers. Redirect budget toward Champions and Loyal Low-Spend to deepen wallet share (r=…

- Loyalty-tier gates: Champions qualify for premium promos only
- Eliminate discount access for Disengaged — test experiential win-back
- Run A/B: value-based (exclusive access) vs discount-based retention

04 MEDIUM Redesign Loyalty Architecture

With 0.01% at Gold tier and 83% at Bronze, the loyalty programme is non-functional as a retention mechanism. Wallet share is the strongest l…

- 3-step progression: Bronze→Silver→Gold with wallet-share milestones
- Tie advancement to wallet_share_percentage, not purchase count
- Subscription-loyalty bridge: premium subscribers fast-track to Silver

05 MEDIUM Investigate Demographic-Behavioural Risk Clusters

Income predicts payment failure (r=–0.49) and stress predicts impulse purchasing (r=+0.74). Neither is embedded in CLV, churn, or marketing …

- Integrate income quintile into credit model: reduce Q1 limits 30–40%
- Flag stress-impulse cluster: high promo sensitivity + high return frequency
- Post-purchase loyalty nudge for impulse buyers to build durable habit

Priority Action Summary

Priority	Action	Owner	Horizon
P1 — Critical	Rebuild CLV formula & rescore base	Analytics / Finance	0–4 wks
P1 — Critical	Value-weight churn model + At-Risk watchlist	Data Science / CRM	4–8 wks
P2 — High	Loyalty-tiered discount gates	Marketing / Product	8–12 wks
P3 — Medium	Loyalty tier architecture redesign	Product / CRM	Q2
P3 — Medium	Demographic risk model integration	Risk / Analytics	Q2–Q3

Fig 6 — Payment & Credit Risk: Income-Stratified Failure Rates · Risk Quadrant

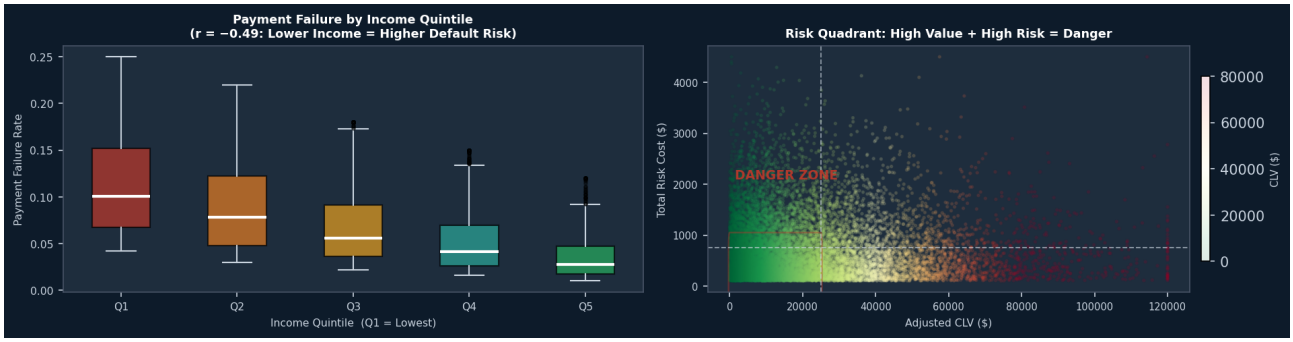


Fig 6 — Q1-income customers show 2.5× higher payment failure rates than Q5 (left); Danger zone identifies customers with high CLV and high risk costs requiring immediate credit-limit review and i